The Global Recovery Gap

Life Expectancy Before, During, and After COVID-19

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Introduction

For my first dataset, I selected life expectancy data from the World Bank. I explored how life expectancy changed across different countries before, during, and after the COVID-19 pandemic. My main goal was to understand whether countries recovered at the same rate or whether some continued to experience lower life expectancy.

Data Source

The dataset comes from the World Bank's World Development Indicators database.

- **Dataset:** Life expectancy at birth, total (years)
- **Indicator:** SP.DYN.LE00.IN
- **Source:** World Bank Life Expectancy Data

The original dataset contains countries, regional aggregates, and separate columns for each year. I also used country metadata to identify regions and income groups.

Research Question

How did life expectancy change between 2019 and 2024, and which countries experienced the strongest and weakest recoveries following the COVID-19 pandemic?

Data Tidying Approach

The original dataset was in wide format, with each year stored in a separate column. I used `pivot_longer()` from the `tidyr` package to convert the year columns into two variables: `Year` and `LifeExpectancy`.

I also used `dplyr` to select the relevant columns, check missing values, and remove regional aggregates using World Bank country metadata. I kept the original CSV files unchanged and completed all transformations in R.

Analysis Approach

I compared life expectancy in 2019, 2021, and 2024 to examine changes during and after the pandemic. I calculated the changes for individual countries and economies and compared patterns across income groups.

I also created line charts to show life expectancy trends and bar charts to highlight countries with the largest declines and strongest recoveries.

Data Challenges

One challenge was separating individual countries and economies from regional aggregates. Some observations also had missing values or unusual year-to-year changes that required further attention.

Another important consideration was that changes in life expectancy could not automatically be attributed entirely to COVID-19, since other factors may have influenced the results.

Project Outcome

The analysis included 217 countries and economies from the World Bank dataset. It identified countries with major life expectancy declines, countries with strong recoveries, and places that remained below their 2019 levels in 2024.

The project also demonstrated how data cleaning, reshaping, joining datasets, and visualization in R can help answer real-world research questions.

Data Preparation and Exploration

Load Required Packages

```
library(tidyverse)
library(knitr)
library(plotly)
library(countrycode)
```

Import the Original Dataset

```r
life_raw <- read_csv(
  "data/life_expectancy_wide.csv",
  skip = 4,
  show_col_types = FALSE
)
```

```
## New names:
## * `` -> `...71`
```

```r
dim(life_raw)
```

```
## [1] 265  71
```

```r
names(life_raw)
```

```
##  [1] "Country Name"   "Country Code"   "Indicator Name" "Indicator Code"
##  [5] "1960"           "1961"           "1962"           "1963"
##  [9] "1964"           "1965"           "1966"           "1967"
## [13] "1968"           "1969"           "1970"           "1971"
## [17] "1972"           "1973"           "1974"           "1975"
## [21] "1976"           "1977"           "1978"           "1979"
## [25] "1980"           "1981"           "1982"           "1983"
## [29] "1984"           "1985"           "1986"           "1987"
## [33] "1988"           "1989"           "1990"           "1991"
## [37] "1992"           "1993"           "1994"           "1995"
## [41] "1996"           "1997"           "1998"           "1999"
## [45] "2000"           "2001"           "2002"           "2003"
## [49] "2004"           "2005"           "2006"           "2007"
## [53] "2008"           "2009"           "2010"           "2011"
## [57] "2012"           "2013"           "2014"           "2015"
## [61] "2016"           "2017"           "2018"           "2019"
## [65] "2020"           "2021"           "2022"           "2023"
## [69] "2024"           "2025"           "...71"
```

```r
head(life_raw)
```

```
## # A tibble: 6 x 71
##   `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
##   <chr>          <chr>          <chr>            <chr>             <dbl>  <dbl>
## 1 Aruba          ABW            Life expectancy~ SP.DYN.LE00.IN     64.0   64.2
## 2 Africa Eastern~ AFE           Life expectancy~ SP.DYN.LE00.IN     44.2   44.5
## 3 Afghanistan    AFG            Life expectancy~ SP.DYN.LE00.IN     32.8   33.3
## 4 Africa Western~ AFW           Life expectancy~ SP.DYN.LE00.IN     37.8   38.1
## 5 Angola         AGO            Life expectancy~ SP.DYN.LE00.IN     37.9   36.9
## 6 Albania        ALB            Life expectancy~ SP.DYN.LE00.IN     56.4   57.5
## # i 65 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
## #   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
## #   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
## #   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
## #   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
## #   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>,
## #   `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>, ...
```

Inspect the Original Data Structure

The original World Bank dataset uses a wide format, where each row represents a country or regional aggregate and each year is stored in a separate column.

```
life_raw %>%
  select(
    `Country Name`,
    `Country Code`,
    `2019`:`2024`
  ) %>%
  glimpse()
```

```
## Rows: 265
## Columns: 8
## $ ‘Country Name‘ <chr> "Aruba", "Africa Eastern and Southern", "Afghanistan", ~
## $ ‘Country Code‘ <chr> "ABW", "AFE", "AFG", "AFW", "AGO", "ALB", "AND", "ARB",~
## $ ‘2019‘         <dbl> 76.01900, 63.85726, 62.94100, 57.14283, 63.05100, 79.46~
## $ ‘2020‘         <dbl> 75.40600, 63.76648, 61.45400, 57.35771, 63.11600, 77.82~
## $ ‘2021‘         <dbl> 73.65500, 62.98000, 60.41700, 57.35502, 62.95800, 76.84~
## $ ‘2022‘         <dbl> 76.22600, 64.48715, 65.61700, 57.97900, 64.24600, 78.76~
## $ ‘2023‘         <dbl> 76.35300, 65.14623, 66.03500, 58.84748, 64.61700, 79.60~
## $ ‘2024‘         <dbl> 76.50000, 65.34993, 66.28900, 59.04154, 64.80500, 79.77~
```

```
life_raw %>%
  select(
    `Country Name`,
    `Country Code`,
    `2019`,
    `2020`,
    `2021`,
    `2022`,
    `2023`,
    `2024`
  ) %>%
  head(10) %>%
  kable(
    caption = "Original Wide-Format Life Expectancy Data"
  )
```

Table 1: Original Wide-Format Life Expectancy Data

Country Name	Country Code	2019	2020	2021	2022	2023	2024
Aruba	ABW	76.01900	75.40600	73.65500	76.22600	76.35300	76.50000
Africa Eastern and Southern	AFE	63.85726	63.76648	62.98000	64.48715	65.14623	65.34993
Afghanistan	AFG	62.94100	61.45400	60.41700	65.61700	66.03500	66.28900
Africa Western and Central	AFW	57.14283	57.35771	57.35502	57.97900	58.84748	59.04154
Angola	AGO	63.05100	63.11600	62.95800	64.24600	64.61700	64.80500
Albania	ALB	79.46700	77.82400	76.84400	78.76900	79.60200	79.77600

Country Name	Country Code	2019	2020	2021	2022	2023	2024
Andorra	AND	84.09800	79.41800	82.33100	84.01600	84.04100	84.18800
Arab World	ARB	71.83989	70.72656	70.40411	71.87610	72.44468	72.72181
United Arab Emirates	ARE	82.59600	81.93600	79.08300	80.48700	82.90900	83.06900
Argentina	ARG	76.84700	75.87800	73.94800	75.80600	77.39500	77.54300

Data Tidying: Wide to Long Format

The original World Bank dataset contained a separate column for each year. I used `pivot_longer()` to transform these columns into a single `Year` variable and a single `LifeExpectancy` variable.

Since my research focused on changes between 2019 and 2024, I selected those years for the analysis.

```
life_tidy <- life_raw %>%
  select(
    `Country Name`,
    `Country Code`,
    `2019`:`2024`
  ) %>%
  pivot_longer(
    cols = `2019`:`2024`,
    names_to = "Year",
    values_to = "LifeExpectancy"
  ) %>%
  mutate(
    Year = as.integer(Year)
  )

head(life_tidy, 12)
```

```
## # A tibble: 12 x 4
##    `Country Name`              `Country Code`  Year LifeExpectancy
##    <chr>                       <chr>          <int>          <dbl>
##  1 Aruba                       ABW             2019           76.0
##  2 Aruba                       ABW             2020           75.4
##  3 Aruba                       ABW             2021           73.7
##  4 Aruba                       ABW             2022           76.2
##  5 Aruba                       ABW             2023           76.4
##  6 Aruba                       ABW             2024           76.5
##  7 Africa Eastern and Southern AFE             2019           63.9
##  8 Africa Eastern and Southern AFE             2020           63.8
##  9 Africa Eastern and Southern AFE             2021           63.0
## 10 Africa Eastern and Southern AFE             2022           64.5
## 11 Africa Eastern and Southern AFE             2023           65.1
## 12 Africa Eastern and Southern AFE             2024           65.3
```

Inspect the Tidied Dataset

```r
cat("Original dataset dimensions:\n")
```

```
## Original dataset dimensions:
```

```r
dim(life_raw)
```

```
## [1] 265  71
```

```r
cat("\nTidied dataset dimensions:\n")
```

```
##
## Tidied dataset dimensions:
```

```r
dim(life_tidy)
```

```
## [1] 1590    4
```

```r
cat("\nTidied dataset structure:\n")
```

```
##
## Tidied dataset structure:
```

```r
glimpse(life_tidy)
```

```
## Rows: 1,590
## Columns: 4
## $ `Country Name` <chr> "Aruba", "Aruba", "Aruba", "Aruba", "Aruba", "Aruba", "~
## $ `Country Code` <chr> "ABW", "ABW", "ABW", "ABW", "ABW", "ABW", "AFE", "AFE",~
## $ Year           <int> 2019, 2020, 2021, 2022, 2023, 2024, 2019, 2020, 2021, 2~
## $ LifeExpectancy <dbl> 76.01900, 75.40600, 73.65500, 76.22600, 76.35300, 76.50~
```

Display the Tidied Data

```r
life_tidy %>%
  head(12) %>%
  kable(
    digits = 2,
    caption = "Life Expectancy Data After Tidying"
  )
```

Table 2: Life Expectancy Data After Tidying

Country Name	Country Code	Year	LifeExpectancy
Aruba	ABW	2019	76.02
Aruba	ABW	2020	75.41

Country Name	Country Code	Year	LifeExpectancy
Aruba	ABW	2021	73.66
Aruba	ABW	2022	76.23
Aruba	ABW	2023	76.35
Aruba	ABW	2024	76.50
Africa Eastern and Southern	AFE	2019	63.86
Africa Eastern and Southern	AFE	2020	63.77
Africa Eastern and Southern	AFE	2021	62.98
Africa Eastern and Southern	AFE	2022	64.49
Africa Eastern and Southern	AFE	2023	65.15
Africa Eastern and Southern	AFE	2024	65.35

Country Metadata and Data Cleaning

The World Bank dataset included both individual countries and regional aggregates. I used the country metadata file to identify countries and economies and add regional and income-group information.

Import Country Metadata

```
country_metadata_raw <- read_csv(
  "data/life_expectancy_countries.csv",
  show_col_types = FALSE
)
```

```
## New names:
## * `` -> `...6`
```

```
dim(country_metadata_raw)
```

```
## [1] 264   6
```

```
names(country_metadata_raw)
```

```
## [1] "Country Code" "Region"       "IncomeGroup"  "SpecialNotes" "TableName"
## [6] "...6"
```

```
country_metadata_raw %>%
  head(10) %>%
  kable(caption = "World Bank Country Metadata")
```

Table 3: World Bank Country Metadata

Country Code	Region	Income Group	Special Notes	Table Name	Note
ABW	Latin America & Caribbean	High income	NA	Aruba	NA
AFE	NA	NA	26 countries, stretching from the Red Sea in the North to the Cape of Good Hope in the South (https://www.worldbank.org/en/region/afr/eastern-and-southern-africa)	Africa Eastern and Southern	NA
AFG	Middle East & North Africa	Low income	The reporting period for national accounts data is designated as either calendar year basis (CY) or fiscal year basis (FY). For this country, it is fiscal year-based (fiscal year-end: March 20). Also, an estimate (PA.NUS.ATLS) of the exchange rate covers the same period and thus differs from the official exchange rate (CY).		

In addition, the World Bank systematically assesses the appropriateness of official exchange rates as conversion factors. In this country, multiple or dual exchange rate activity exists and must be accounted for appropriately in underlying statistics. An alternative estimate ("alternative conversion factor" - PA.NUS.ATLS) is thus calculated as a weighted average of the different exchange rates in use in the country. Doing so better reflects economic reality and leads to more accurate cross-country comparisons and country classifications by income level. For this country, this applies to the period 1960-2006. Alternative conversion factors are used in the Atlas methodology and elsewhere in World Development Indicators as single-year conversion factors. |Afghanistan |NA | |AFW |NA |NA |22 countries, stretching from the westernmost point of Africa, across the equator, and partly along the Atlantic Ocean till the Republic of Congo in the South (https://www.worldbank.org/en/region/afr/western-and-central-africa) |Africa Western and Central |NA | |AGO |Sub-Saharan Africa |Lower middle income |The World Bank systematically assesses the appropriateness of official exchange rates as conversion factors. In this country, multiple or dual exchange rate activity exists and must be accounted for appropriately in underlying statistics. An alternative estimate ("alternative conversion factor" - PA.NUS.ATLS) is thus calculated as a weighted average of the different exchange rates in use in the country. Doing so better reflects economic reality and leads to more accurate cross-country comparisons and country classifications by income level. For this country, this applies to the period 1994-2025. Alternative conversion factors are used in the Atlas methodology and elsewhere in World Development Indicators as single-year conversion factors. |Angola |NA | |ALB |Europe & Central Asia |Upper middle income |NA |Albania |NA | |AND |Europe & Central Asia |High income |NA |Andorra |NA | |ARB |NA |NA |Arab World aggregate. Arab World is composed of members of the League of Arab States. |Arab World |NA | |ARE |Middle East & North Africa |High income |NA |United Arab Emirates |NA | |ARG |Latin America & Caribbean |Upper middle income |Due to the limitations of the Atlas method in classifying countries experiencing persistent high inflation, this economy remains classified as upper-middle-income. The World Bank systematically assesses the appropriateness of official exchange rates as conversion factors. In this country, multiple or dual exchange rate activity exists and must be accounted for appropriately in underlying statistics. An alternative estimate ("alternative conversion factor" - PA.NUS.ATLS) is thus calculated as a weighted average of the different exchange rates in use in the country. Doing so better reflects economic reality and leads to more accurate cross-country comparisons and country classifications by income level. For this country, this applies to the period 1971-2018. Alternative conversion factors are used in the Atlas methodology and elsewhere in World Development Indicators as single-year conversion factors. |Argentina |NA |

Inspect Metadata Structure

```
glimpse(country_metadata_raw)
```

```
## Rows: 264
## Columns: 6
## $ 'Country Code' <chr> "ABW", "AFE", "AFG", "AFW", "AGO", "ALB", "AND", "ARB",~
## $ Region         <chr> "Latin America & Caribbean", NA, "Middle East & North A~
## $ IncomeGroup    <chr> "High income", NA, "Low income", NA, "Lower middle inco~
## $ SpecialNotes   <chr> NA, "26 countries, stretching from the Red Sea in the N~
## $ TableName      <chr> "Aruba", "Africa Eastern and Southern", "Afghanistan", ~
## $ ...6           <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
```

Joining and Cleaning the Data

I joined the tidied life expectancy dataset with the World Bank country metadata using `Country Code`. I then removed regional aggregates by keeping records with valid region and income-group classifications.

Prepare Country Metadata

```
country_metadata <- country_metadata_raw %>%
  select(
    `Country Code`,
    Region,
    IncomeGroup
  ) %>%
  distinct(`Country Code`, .keep_all = TRUE)

country_metadata %>%
  head(10) %>%
  kable(caption = "Cleaned Country Metadata")
```

Table 4: Cleaned Country Metadata

Country Code	Region	IncomeGroup
ABW	Latin America & Caribbean	High income
AFE	NA	NA
AFG	Middle East & North Africa	Low income
AFW	NA	NA
AGO	Sub-Saharan Africa	Lower middle income
ALB	Europe & Central Asia	Upper middle income
AND	Europe & Central Asia	High income
ARB	NA	NA
ARE	Middle East & North Africa	High income
ARG	Latin America & Caribbean	Upper middle income

Join Life Expectancy and Country Metadata

```
life_joined <- life_tidy %>%
  left_join(
    country_metadata,
    by = "Country Code"
  )

cat("Rows after joining:", nrow(life_joined), "\n")
```

```
## Rows after joining: 1590
```

```
life_joined %>%
  head(10) %>%
  kable(
    digits = 2,
    caption = "Life Expectancy Data with Country Metadata"
  )
```

Table 5: Life Expectancy Data with Country Metadata

Country Name	Country Code	Year	LifeExpectancy	Region	IncomeGroup
Aruba	ABW	2019	76.02	Latin America & Caribbean	High income
Aruba	ABW	2020	75.41	Latin America & Caribbean	High income
Aruba	ABW	2021	73.66	Latin America & Caribbean	High income
Aruba	ABW	2022	76.23	Latin America & Caribbean	High income
Aruba	ABW	2023	76.35	Latin America & Caribbean	High income
Aruba	ABW	2024	76.50	Latin America & Caribbean	High income
Africa Eastern and Southern	AFE	2019	63.86	NA	NA
Africa Eastern and Southern	AFE	2020	63.77	NA	NA
Africa Eastern and Southern	AFE	2021	62.98	NA	NA
Africa Eastern and Southern	AFE	2022	64.49	NA	NA

Remove Regional Aggregates

```
life_clean <- life_joined %>%
  filter(
    !is.na(Region),
```

```
    !is.na(IncomeGroup)
  )

cat("Rows before filtering:", nrow(life_joined), "\n")
```

```
## Rows before filtering: 1590
```

```
cat("Rows after filtering:", nrow(life_clean), "\n")
```

```
## Rows after filtering: 1302
```

```
cat(
  "Number of countries:",
  n_distinct(life_clean$`Country Code`),
  "\n"
)
```

```
## Number of countries: 217
```

```
life_clean %>%
  head(10) %>%
  kable(
    digits = 2,
    caption = "Cleaned Country-Level Life Expectancy Data"
  )
```

Table 6: Cleaned Country-Level Life Expectancy Data

Country Name	Country Code	Year	LifeExpectancy	Region	IncomeGroup
Aruba	ABW	2019	76.02	Latin America & Caribbean	High income
Aruba	ABW	2020	75.41	Latin America & Caribbean	High income
Aruba	ABW	2021	73.66	Latin America & Caribbean	High income
Aruba	ABW	2022	76.23	Latin America & Caribbean	High income
Aruba	ABW	2023	76.35	Latin America & Caribbean	High income
Aruba	ABW	2024	76.50	Latin America & Caribbean	High income
Afghanistan	AFG	2019	62.94	Middle East & North Africa	Low income
Afghanistan	AFG	2020	61.45	Middle East & North Africa	Low income
Afghanistan	AFG	2021	60.42	Middle East & North Africa	Low income
Afghanistan	AFG	2022	65.62	Middle East & North Africa	Low income

Check Missing Values

```
missing_summary <- life_clean %>%
  group_by(Year) %>%
  summarise(
    TotalRecords = n(),
    MissingLifeExpectancy = sum(is.na(LifeExpectancy)),
```

```
    AvailableLifeExpectancy = sum(!is.na(LifeExpectancy)),
    .groups = "drop"
  )

missing_summary %>%
  kable(
    caption = "Missing Life Expectancy Values by Year"
  )
```

Table 7: Missing Life Expectancy Values by Year

Year	TotalRecords	MissingLifeExpectancy	AvailableLifeExpectancy
2019	217	0	217
2020	217	0	217
2021	217	0	217
2022	217	0	217
2023	217	0	217
2024	217	0	217

Life Expectancy Decline and Recovery Analysis

To understand how life expectancy changed during and after the COVID-19 pandemic, I compared life expectancy in 2019, 2021, and 2024.

The analysis measured changes during the pandemic period, recovery after 2021, and the overall difference from the pre-pandemic baseline.

Prepare Country-Level Comparisons

```
life_changes <- life_clean %>%
  filter(Year %in% c(2019, 2021, 2024)) %>%
  select(
    `Country Name`,
    `Country Code`,
    Region,
    IncomeGroup,
    Year,
    LifeExpectancy
  ) %>%
  pivot_wider(
    names_from = Year,
    values_from = LifeExpectancy,
    names_prefix = "Life_"
  ) %>%
  mutate(
    PandemicChange = Life_2021 - Life_2019,
    RecoveryChange = Life_2024 - Life_2021,
    OverallChange = Life_2024 - Life_2019
  )
```

```
life_changes %>%
  head(10) %>%
  kable(
    digits = 2,
    caption = "Life Expectancy Changes by Country"
  )
```

Table 8: Life Expectancy Changes by Country

Country Name	Country Code	Region	IncomeGroup	Life_2019	Life_2021	Life_2024	PandemicChange	RecoveryChange	OverallChange
Aruba	ABW	Latin America & Caribbean	High income	76.02	73.66	76.50	-2.36	2.84	0.48
Afghanistan	AFG	Middle East & North Africa	Low income	62.94	60.42	66.29	-2.52	5.87	3.35
Angola	AGO	Sub-Saharan Africa	Lower middle income	63.05	62.96	64.81	-0.09	1.85	1.75
Albania	ALB	Europe & Central Asia	Upper middle income	79.47	76.84	79.78	-2.62	2.93	0.31
Andorra	AND	Europe & Central Asia	High income	84.10	82.33	84.19	-1.77	1.86	0.09
United Arab Emirates	ARE	Middle East & North Africa	High income	82.60	79.08	83.07	-3.51	3.99	0.47
Argentina	ARG	Latin America & Caribbean	Upper middle income	76.85	73.95	77.54	-2.90	3.60	0.70
Armenia	ARM	Europe & Central Asia	Upper middle income	76.22	72.28	78.32	-3.94	6.04	2.10
American Samoa	ASM	East Asia & Pacific	High income	72.75	72.71	72.99	-0.04	0.28	0.24
Antigua and Barbuda	ATG	Latin America & Caribbean	High income	77.17	77.20	77.77	0.03	0.57	0.60

Check Available Comparisons

```
life_changes %>%
  summarise(
    TotalCountries = n(),
    CompleteComparisons = sum(
      !is.na(Life_2019) &
      !is.na(Life_2021) &
      !is.na(Life_2024)
    ),
    MissingComparisons = sum(
```

```
      is.na(Life_2019) |
      is.na(Life_2021) |
      is.na(Life_2024)
    )
  ) %>%
  kable(
    caption = "Data Availability for Country Comparisons"
  )
```

Table 9: Data Availability for Country Comparisons

TotalCountries	CompleteComparisons	MissingComparisons
217	217	0

Countries with the Largest Declines

```
life_changes %>%
  filter(!is.na(PandemicChange)) %>%
  arrange(PandemicChange) %>%
  select(
    `Country Name`,
    Region,
    PandemicChange
  ) %>%
  head(10) %>%
  kable(
    digits = 2,
    caption = "10 Largest Life Expectancy Declines, 2019-2021"
  )
```

Table 10: 10 Largest Life Expectancy Declines, 2019–2021

Country Name	Region	PandemicChange
Bolivia	Latin America & Caribbean	-6.39
Paraguay	Latin America & Caribbean	-5.56
Mexico	Latin America & Caribbean	-4.78
Guyana	Latin America & Caribbean	-4.74
Peru	Latin America & Caribbean	-4.68
Lebanon	Middle East & North Africa	-4.56
Ecuador	Latin America & Caribbean	-4.54
Cuba	Latin America & Caribbean	-4.21
Colombia	Latin America & Caribbean	-4.10
South Africa	Sub-Saharan Africa	-4.06

Countries with the Strongest Recoveries

```r
life_changes %>%
  filter(!is.na(RecoveryChange)) %>%
  arrange(desc(RecoveryChange)) %>%
  select(
    `Country Name`,
    Region,
    RecoveryChange
  ) %>%
  head(10) %>%
  kable(
    digits = 2,
    caption = "10 Largest Life Expectancy Increases, 2021-2024"
  )
```

Table 11: 10 Largest Life Expectancy Increases, 2021–2024

Country Name	Region	RecoveryChange
Central African Republic	Sub-Saharan Africa	17.39
Bolivia	Latin America & Caribbean	7.31
Namibia	Sub-Saharan Africa	6.67
Peru	Latin America & Caribbean	6.34
Armenia	Europe & Central Asia	6.04
Eswatini	Sub-Saharan Africa	6.04
Guyana	Latin America & Caribbean	6.00
Botswana	Sub-Saharan Africa	5.99
Paraguay	Latin America & Caribbean	5.87
Afghanistan	Middle East & North Africa	5.87

Countries Still Below Their 2019 Levels

```r
life_changes %>%
  filter(!is.na(OverallChange), OverallChange < 0) %>%
  arrange(OverallChange) %>%
  select(
    `Country Name`,
    Region,
    OverallChange
  ) %>%
  head(10) %>%
  kable(
    digits = 2,
    caption = "Countries Furthest Below 2019 Life Expectancy in 2024"
  )
```

Table 12: Countries Furthest Below 2019 Life Expectancy in 2024

Country Name	Region	OverallChange
West Bank and Gaza	Middle East & North Africa	-6.60

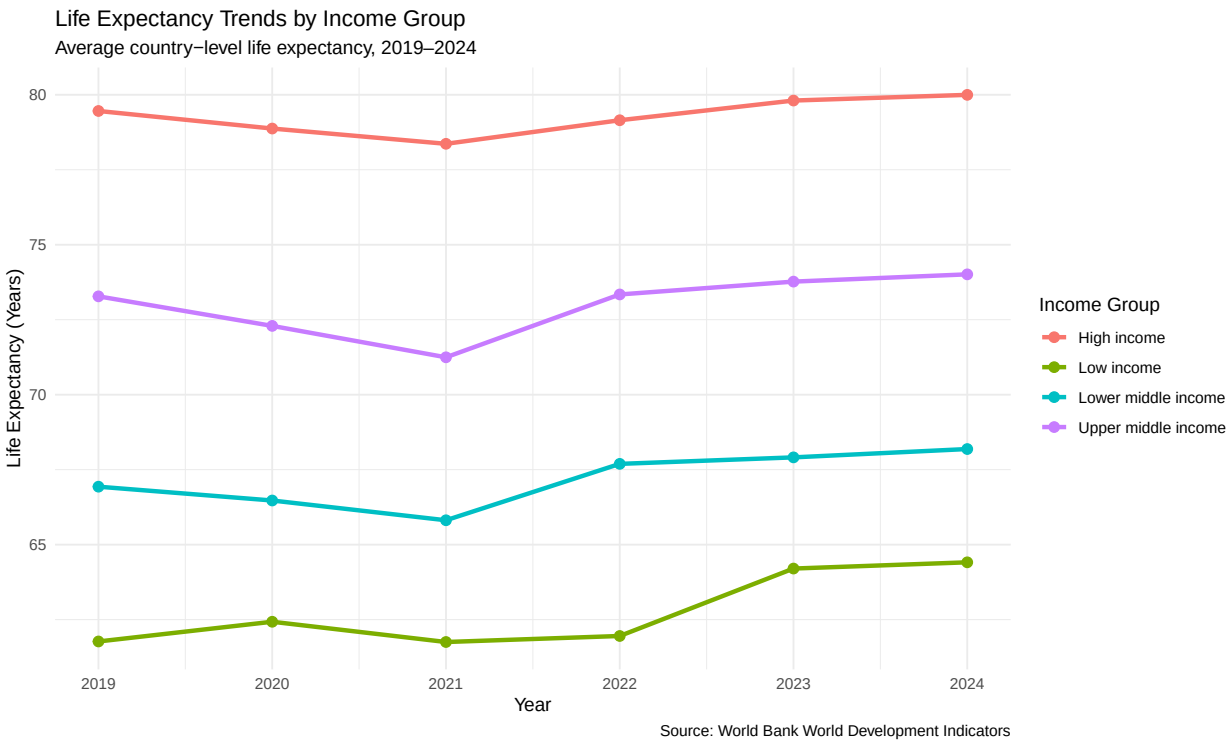
Country Name	Region	OverallChange
Libya	Middle East & North Africa	-1.82
Greenland	Europe & Central Asia	-1.14
Thailand	East Asia & Pacific	-0.63
Macao SAR, China	East Asia & Pacific	-0.60
Germany	Europe & Central Asia	-0.50
Mauritius	Sub-Saharan Africa	-0.45
Qatar	Middle East & North Africa	-0.42
South Sudan	Sub-Saharan Africa	-0.39
Iceland	Europe & Central Asia	-0.35

Data Visualization

Life Expectancy Trends by Income Group

```
income_trends <- life_clean %>%
  group_by(IncomeGroup, Year) %>%
  summarise(
    AverageLifeExpectancy = mean(
      LifeExpectancy,
      na.rm = TRUE
    ),
    .groups = "drop"
  )

ggplot(
  income_trends,
  aes(
    x = Year,
    y = AverageLifeExpectancy,
    color = IncomeGroup,
    group = IncomeGroup
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2.5) +
  scale_x_continuous(breaks = 2019:2024) +
  labs(
    title = "Life Expectancy Trends by Income Group",
    subtitle = "Average country-level life expectancy, 2019-2024",
    x = "Year",
    y = "Life Expectancy (Years)",
    color = "Income Group",
    caption = "Source: World Bank World Development Indicators"
  ) +
  theme_minimal()
```



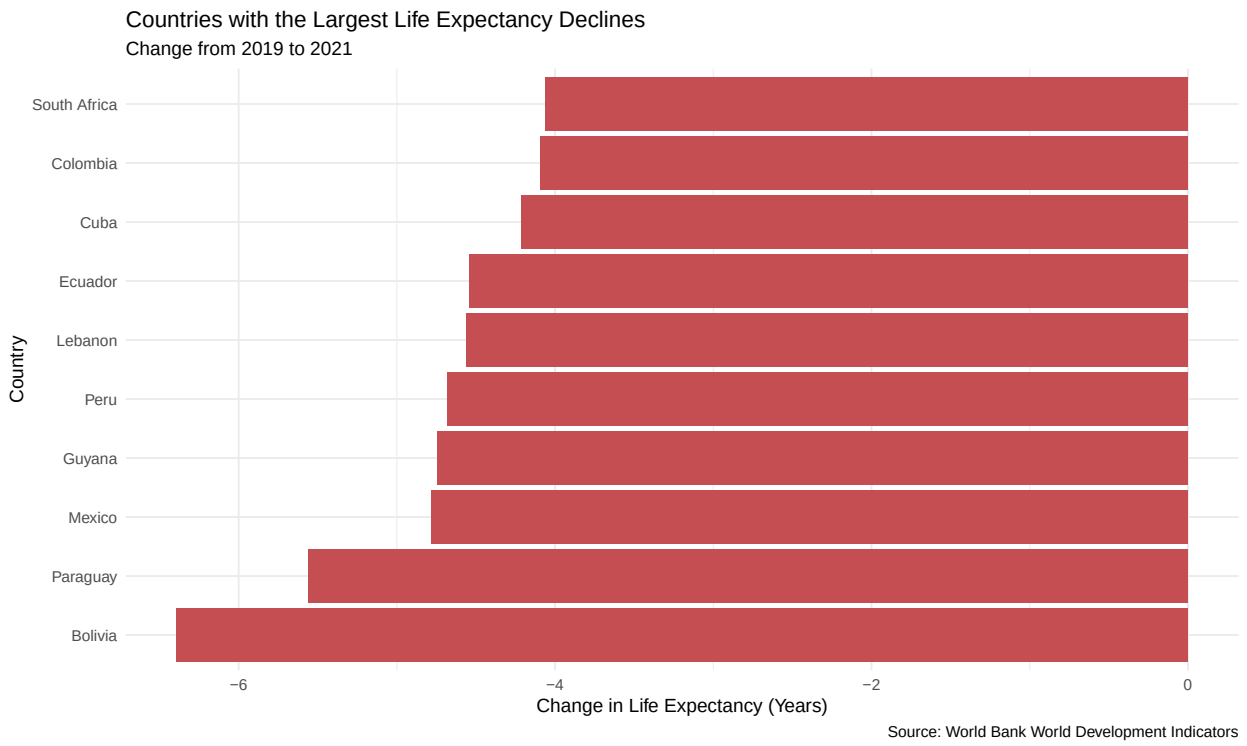
This visualization compares average life expectancy across different income groups from 2019 to 2024. It helps show how life expectancy changed during and after the COVID-19 pandemic and whether recovery patterns differed between income groups. The chart also highlights the differences in life expectancy levels across these groups. Each average represents the countries and economies with available data, rather than a population-weighted global average.

Largest Life Expectancy Declines

```
largest_declines <- life_changes %>%
  filter(!is.na(PandemicChange)) %>%
  slice_min(
    PandemicChange,
    n = 10,
    with_ties = FALSE
  )

ggplot(
  largest_declines,
  aes(
    x = reorder(`Country Name`, PandemicChange),
    y = PandemicChange
  )
) +
  geom_col(fill = "#C44E52") +
  coord_flip() +
  labs(
    title = "Countries with the Largest Life Expectancy Declines",
    subtitle = "Change from 2019 to 2021",
```

```
    x = "Country",
    y = "Change in Life Expectancy (Years)",
    caption = "Source: World Bank World Development Indicators"
  ) +
  theme_minimal()
```



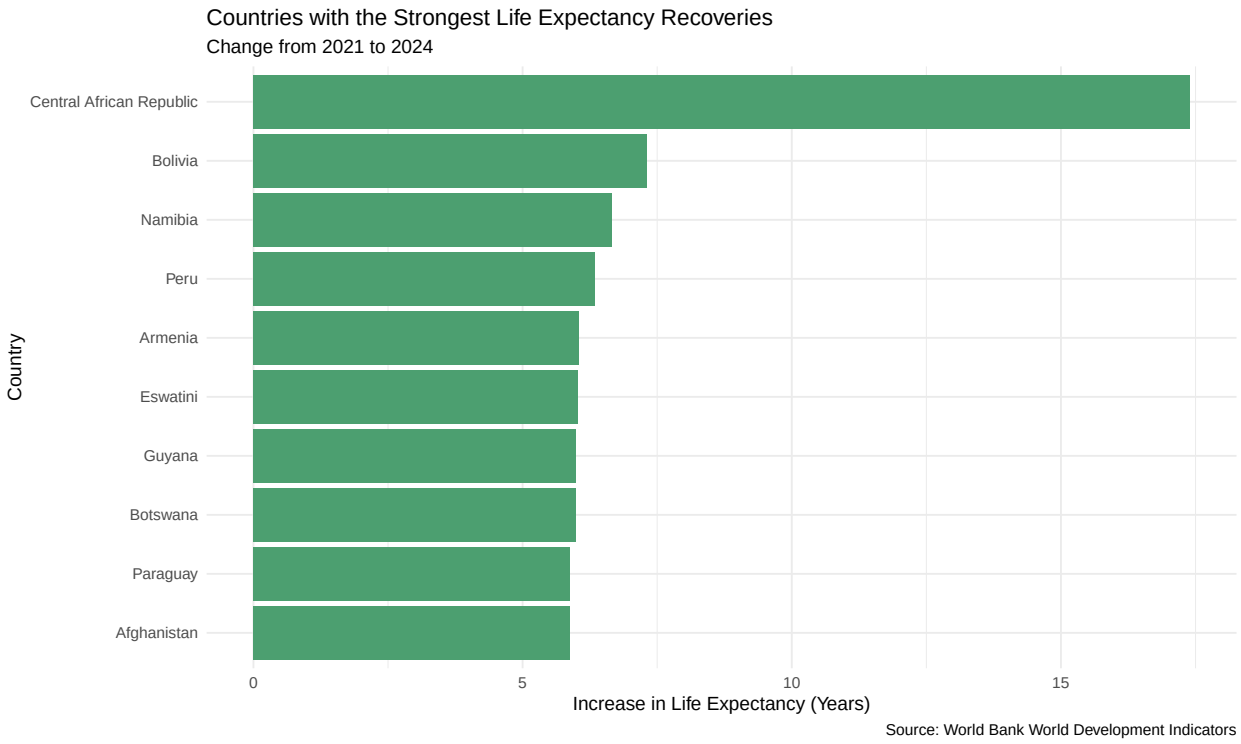
This visualization shows the 10 countries with the largest life expectancy declines between 2019 and 2021. Bolivia experienced the largest decline at approximately 6.39 years, followed by Paraguay at 5.56 years and Mexico at 4.78 years. Several Latin American countries appear in this chart, showing how strongly life expectancy was affected in parts of the region during the pandemic period.

Strongest Life Expectancy Recoveries

```
strongest_recoveries <- life_changes %>%
  filter(!is.na(RecoveryChange)) %>%
  slice_max(
    RecoveryChange,
    n = 10,
    with_ties = FALSE
  )

ggplot(
  strongest_recoveries,
  aes(
    x = reorder(`Country Name`, RecoveryChange),
    y = RecoveryChange
  )
```

```
) +
  geom_col(fill = "#4C9F70") +
  coord_flip() +
  labs(
    title = "Countries with the Strongest Life Expectancy Recoveries",
    subtitle = "Change from 2021 to 2024",
    x = "Country",
    y = "Increase in Life Expectancy (Years)",
    caption = "Source: World Bank World Development Indicators"
  ) +
  theme_minimal()
```



This visualization shows the countries with the largest increases in life expectancy between 2021 and 2024. The Central African Republic had the largest reported increase at 17.4 years, but its data contain unusual fluctuations that require verification. Bolivia followed with an increase of 7.31 years, Namibia with 6.67 years, and Peru with 6.34 years. The results show that countries recovered at different rates, and a strong recovery did not always mean returning to pre-pandemic levels.

Data Quality Check: Central African Republic

The Central African Republic has unusually large changes in life expectancy between consecutive years. I reviewed these observations separately because they may influence the ranking of countries with the strongest recoveries.

```
life_clean %>%
  filter(`Country Code` == "CAF") %>%
  arrange(Year) %>%
  select(Year, LifeExpectancy) %>%
```

```
  kable(
    digits = 2,
    caption = "Central African Republic Life Expectancy, 2019-2024"
  )
```

Table 13: Central African Republic Life Expectancy, 2019–2024

Year	LifeExpectancy
2019	31.53
2020	50.60
2021	40.28
2022	18.82
2023	57.41
2024	57.67

These unusually large changes, especially the decrease in 2022 followed by a sharp increase in 2023, suggest that this country's results should be interpreted cautiously until the source data can be verified.

Interactive World Life Expectancy Recovery Map

This interactive map shows how life expectancy changed across countries between 2019 and 2024 compared with their 2019 levels. Use the year slider to explore changes over time.

```
map_data <- life_clean %>%
  group_by(`Country Code`) %>%
  mutate(
    Life2019 = LifeExpectancy[match(2019, Year)],
    ChangeFrom2019 = LifeExpectancy - Life2019
  ) %>%
  ungroup() %>%
  filter(
    !is.na(ChangeFrom2019),
    !is.na(LifeExpectancy)
  ) %>%
  mutate(
    HoverText = paste0(
      "<b>", `Country Name`, "</b>",
      "<br>Year: ", Year,
      "<br>Life Expectancy: ",
      round(LifeExpectancy, 2), " years",
      "<br>Change from 2019: ",
      sprintf("%+.2f", ChangeFrom2019), " years"
    )
  )

plot_ly(
  data = map_data,
  type = "choropleth",
  locations = ~`Country Code`,
  locationmode = "ISO-3",
  z = ~ChangeFrom2019,
```

```
  frame = ~Year,
  text = ~HoverText,
  hoverinfo = "text",
  colorscale = list(
    c(0, "#C44E52"),
    c(0.5, "#F7F7F7"),
    c(1, "#4C9F70")
  ),
  zmin = -10,
  zmax = 10,
  zmid = 0,
  colorbar = list(title = "Change (Years)")
) %>%
  layout(
    title = "Global Life Expectancy Change Since 2019",
    geo = list(
      showframe = FALSE,
      showcoastlines = TRUE,
      projection = list(type = "natural earth")
    )
  ) %>%
  animation_opts(
    frame = 1000,
    transition = 500,
    redraw = TRUE
  ) %>%
  animation_slider(
    currentvalue = list(prefix = "Year: ")
  )
```

The interactive map uses red to represent countries with lower life expectancy than in 2019 and green for countries with higher life expectancy. Viewers can move the year slider to compare different years and hover over countries to see their values. Countries without available data or a matching map location may not be colored.

Results and Discussion

Countries Still Below Their 2019 Levels

Although many countries experienced increases in life expectancy after 2021, some had not returned to their 2019 levels by 2024. West Bank and Gaza showed the largest overall decline at approximately 6.60 years, followed by Libya and Greenland. This shows that recovery was not equal across countries.

Limitations

Some countries have missing data or unusual changes that require verification. The dataset also includes territories and economies, not just independent countries. Additionally, changes in life expectancy cannot be attributed entirely to COVID-19 because other factors may have influenced the results. Countries with missing life expectancy values were excluded from comparisons requiring those years, so the rankings only include countries with available data.

Conclusion

This project analyzed life expectancy across 217 countries and economies from 2019 to 2024 using World Bank data. The results show that many countries experienced declines during the pandemic period and recovered afterward, but not at the same rate. By cleaning, transforming, and visualizing the data in R, I was able to identify differences in life expectancy recovery across countries and income groups.